Varendra University

Dept. of Computer Science & Engineering (CSE)

Logic, Shift and Rotate Instructions (Chapter-7)

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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Outline

1. Logic Instruction.
2. AND, OR, XOR Instruction.
3. Converting an ASCII Digit to a Number.
4. Converting a Lowercase to Upper Case.
5. Clearing a Register.
6. Testing a Register for Zero.
7. NOT Instruction.
8. TEST Instruction.
9. Shift Instruction.
10. Rotate Instruction.
11. Application of Shift and Rotate Instruction.

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Logic Instruction

Example 7.1 Perform the following logic operations:

1. 10101010 AND 11110000
2. 10101010 OR 11110000
3. 10101010 XOR 11110000
4. NOT 10101010

Solutions:

1.
```
  10101010
AND 11110000
------------
= 10100000
```

2.
```
  10101010
OR 11110000
------------
= 11111010
```

3.
```
  10101010
XOR 11110000
------------
= 01011010
```

4.
```
NOT 10101010
= 01010101
```

Logic Instruction

Figure 7.1 Truth Tables for AND, OR, XOR, and NOT (0 = false, 1 = true)

a	b	a AND b	a OR b	a XOR b
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

a	NOT a
0	1
1	0

AND, OR, XOR Instruction

The **AND**, **OR**, and **XOR** instructions perform the named logic operations. The formats are

```
AND     destination,source
OR      destination,source
XOR     destination,source
```

The result of the operation is stored in the destination, which must be a register or memory location. The source may be a constant, register, or memory location. However, memory-to-memory operations are not allowed.

Effect on flags:

SF, ZF, PF reflect the result

AF is undefined

CF, OF = 0

Converting an ASCII Digit to a Number

when a program reads a character from the keyboard, AL gets the ASCII code of the character. This is also true of digit characters. For example, if the "5" key is pressed, AL gets 35h instead of 5. To get 5 in AL, we could do this:

```
SUB     AL, 30h
```

Another method is to use the AND instruction to clear the high nibble (high four bits) of AL:

```
AND     AL, 0Fh
```

Because the codes of "0" to "9" are 30h to 39h, this method will convert any ASCII digit to a decimal value.

Converting a Lowercase to Upper Case

- The ASCII codes of "a" to "z" range from 61h to 7Ah.
- The ASCII codes of "A" to "Z" range from 41h to 5Ah.

Clearing a Register

We already know two ways to clear a register. For example, to clear AX we could execute

```
MOV     AX, 0
```

or

```
SUB     AX, AX
```

Using the fact that 1 XOR 1 = 0 and 0 XOR 0 = 0, a third way is

```
XOR     AX, AX
```

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Testing a Register for Zero

Because 1 OR 1 = 1, 0 OR 0 = 0, it might seem like a waste of time to execute an instruction like

```
OR      CX,CX
```

because it leaves the contents of CX unchanged. However, it affects ZF and SF, and in particular if CX contains 0 then ZF = 1. So it can be used as an alternative to

```
CMP     CX,0
```

to test the contents of a register for zero, or to check the sign of the contents.

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NOT Instruction

The **NOT** instruction performs the one's complement operation on the destination. The format is

```
NOT     destination
```

There is no effect on the status flags.

Example 7.5 Complement the bits in AX.

Solution:

```
NOT     AX
```

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TEST Instruction

The TEST instruction performs an AND operation of the destination with the source but does not change the destination contents. The purpose of the TEST instruction is to set the status flags.

The format is,

```
TEST destination,source
```

Effect on flags

SF, ZF, PF reflect the result
AF is undefined
CF, OF = 0

Shift Instruction

- The shift and rotate instructions shift the bits in the destination operand by one or more positions either to the left or right.
- For a shift instruction, the bits shifted out are lost.
- For a rotate instruction, bits shifted out from one end of the operand are put back into the other end.

For a single shift rotate, the format is,

```
Opcode    destination,1
```

For a shift or rotate of N positions, the format is,

```
Opcode    destination,CL
```

Where CL contains N.

Shift Instruction

Left Shift Instruction:

The SHL (shift left) instruction shifts the bits in the destination to the left. The format for a single shift is,

```
SHL    destination,1
```

If the shift count N, the format is,

```
SHL    destination,CL
```

Where CL contains N.

Effect on flags

SF, PF, ZF reflect the result

AF is undefined

CF = last bit shifted out

OF = 1 if result changes sign on last shift

Shift Instruction

Left Shift Instruction:

Example 7.7: Suppose DH contains 8Ah and CL contains 3. What are the values of DH and of CF after the instruction SHL DH,CL is executed?

Solution:

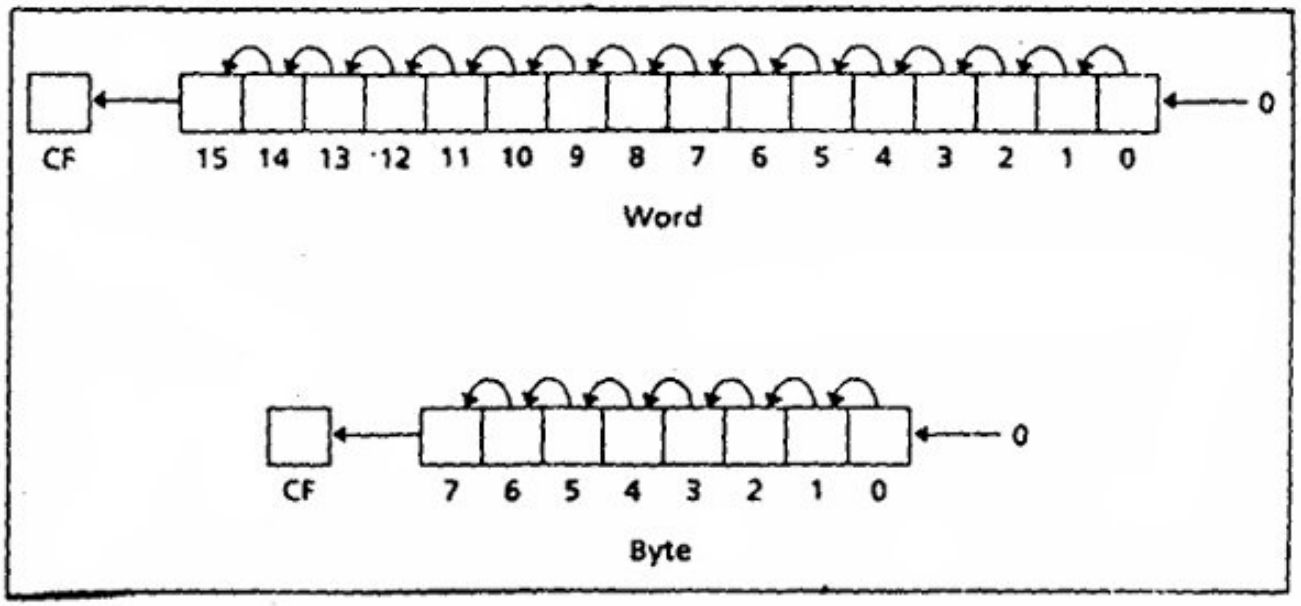
```
MOV DH, 8Ah
MOV CL, 3
SHL DH, CL
```

The **binary value** of **DH is 10001010**. After **3 left shifts**, CF will contain 0. The new contents of DH may be obtained by **erasing the leftmost three bits and adding three zero bits to the right** end, thus **01010000b = 50h**.

Shift Instruction

Left Shift Instruction:

Figure 7.2 SHL



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Shift Instruction

Right Shift Instruction:

The SHR (shift right) instruction shifts the bits in the destination to the right. The format for a single shift is,

SHR destination, 1

If the shift count N, the format is,

```
SHR    destination,CL
```

Where CL contains N.

Effect on flags

SF, PF, ZF reflect the result

AF is undefined

CF = last bit shifted out

OF = 1 if result changes sign on last shift

Shift Instruction

Right Shift Instruction:

Example 7.9: Suppose DH contains 8Ah and CL contains 2. What are the values of DH and of CF after the instruction SHR DH,CL is executed?

Solution:

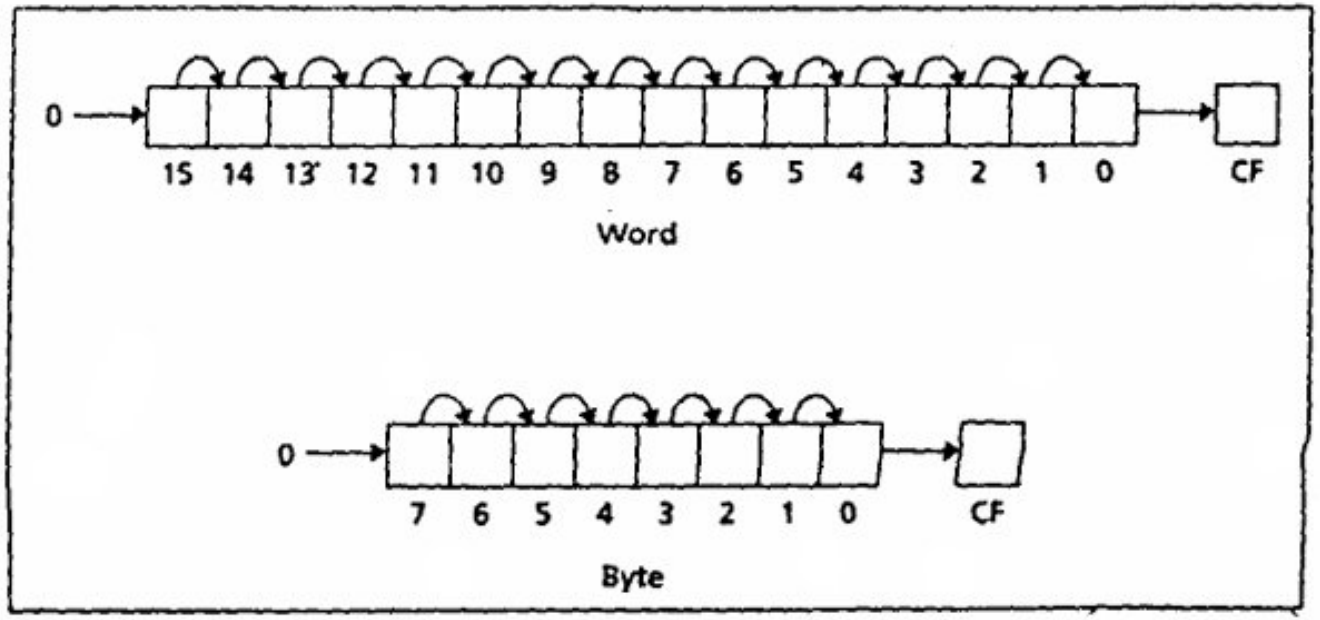
MOV DH, 8Ah
MOV CL, 2
SHR DH, CL

The **binary value** of **DH is 10001010**. After **2 right shifts**, CF = 1. The new contents of DH may be obtained by **erasing the rightmost two bits and adding two zero bits to the left** end, thus **00100010b = 22h**.

Shift Instruction

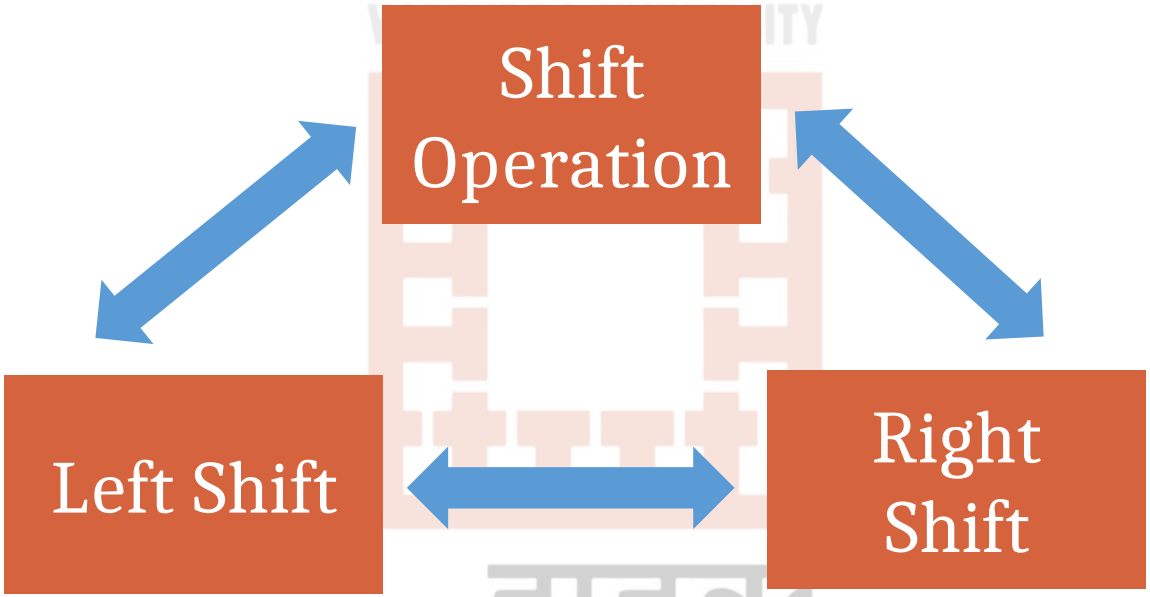
Right Shift Instruction:

Figure 7.3 SHR



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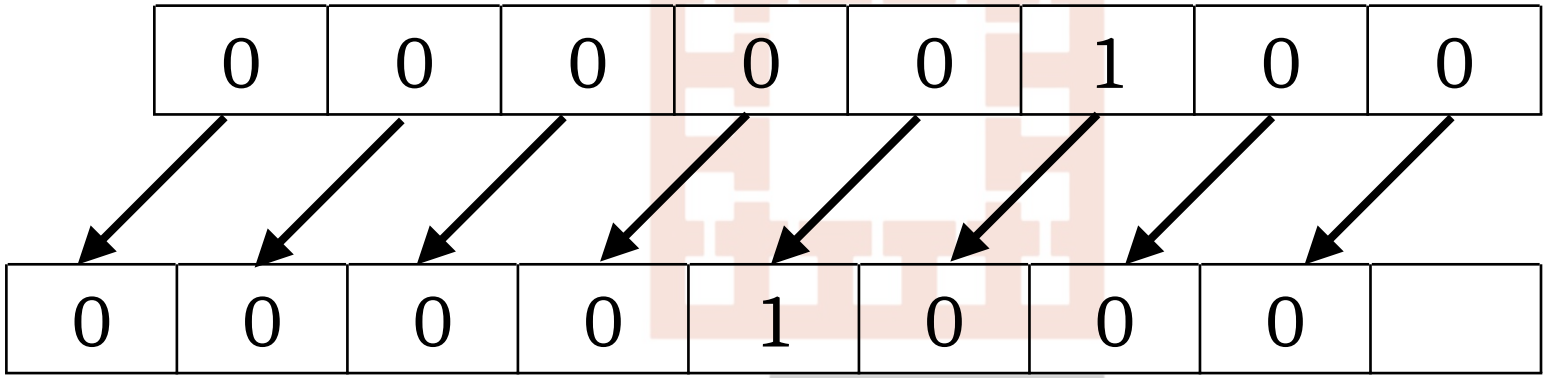
Shift Instruction



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Shift Instruction

Left Shift : Multiply a number by 2
Right Shift : Divide a number by 2



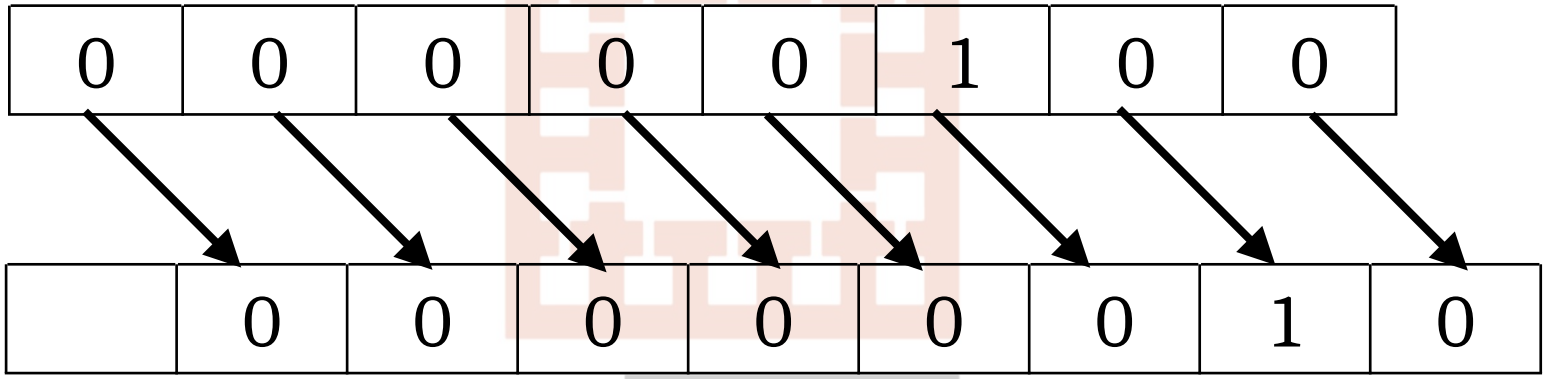
4*2 = 8

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Shift Instruction

Left Shift : Multiply a number by 2

Right Shift : Divide a number by 2



4/2 = 2

Rotate Instruction

Rotate Left:

The instruction ROL (rotate left) shifts bits to the left. The MSB is shifted into the rightmost bit. The CF also gets the bit shifted out of the MSB. The syntax is,

```
ROL     destination,1

and

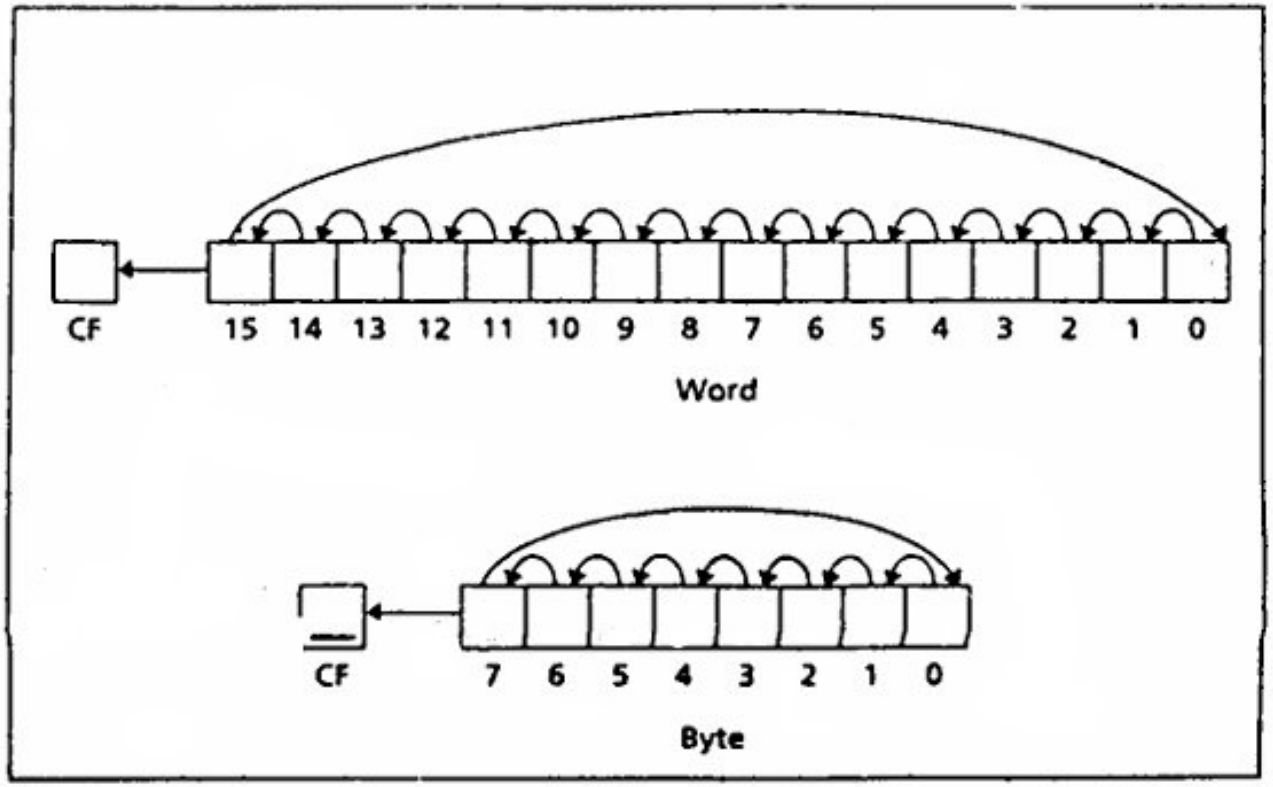
ROL     destination,CL
```

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Rotate Instruction

Rotate Left:

Figure 7.5 ROL



Rotate Instruction

Rotate Right:

The instruction ROR (rotate right) works just like ROL, except that the bits are rotated to the right. The rightmost bit is shifted into the MSB, and also into the CF. The syntax is,

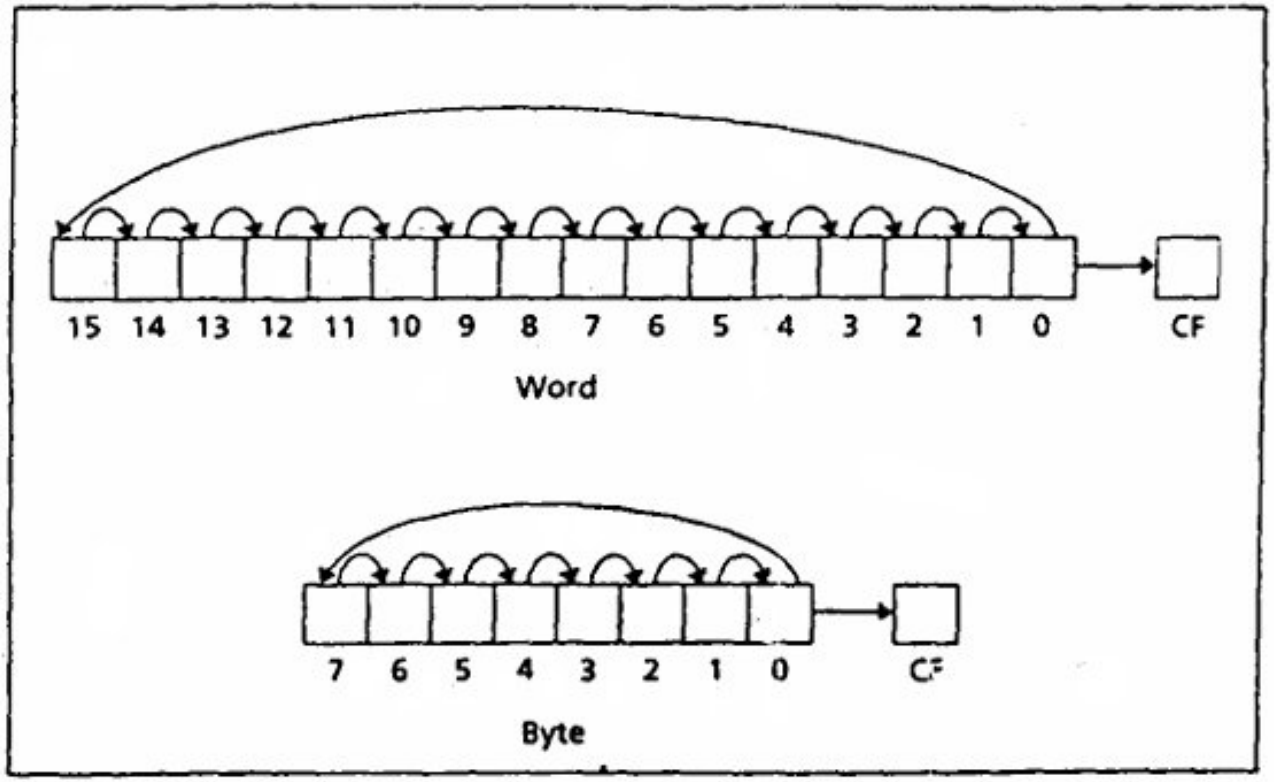
```
ROR     destination,1
and
ROR     destination,CL
```

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Rotate Instruction

Rotate Right:

Figure 7.6 ROR



Rotate Instruction

Rotate Carry Left (RCL):

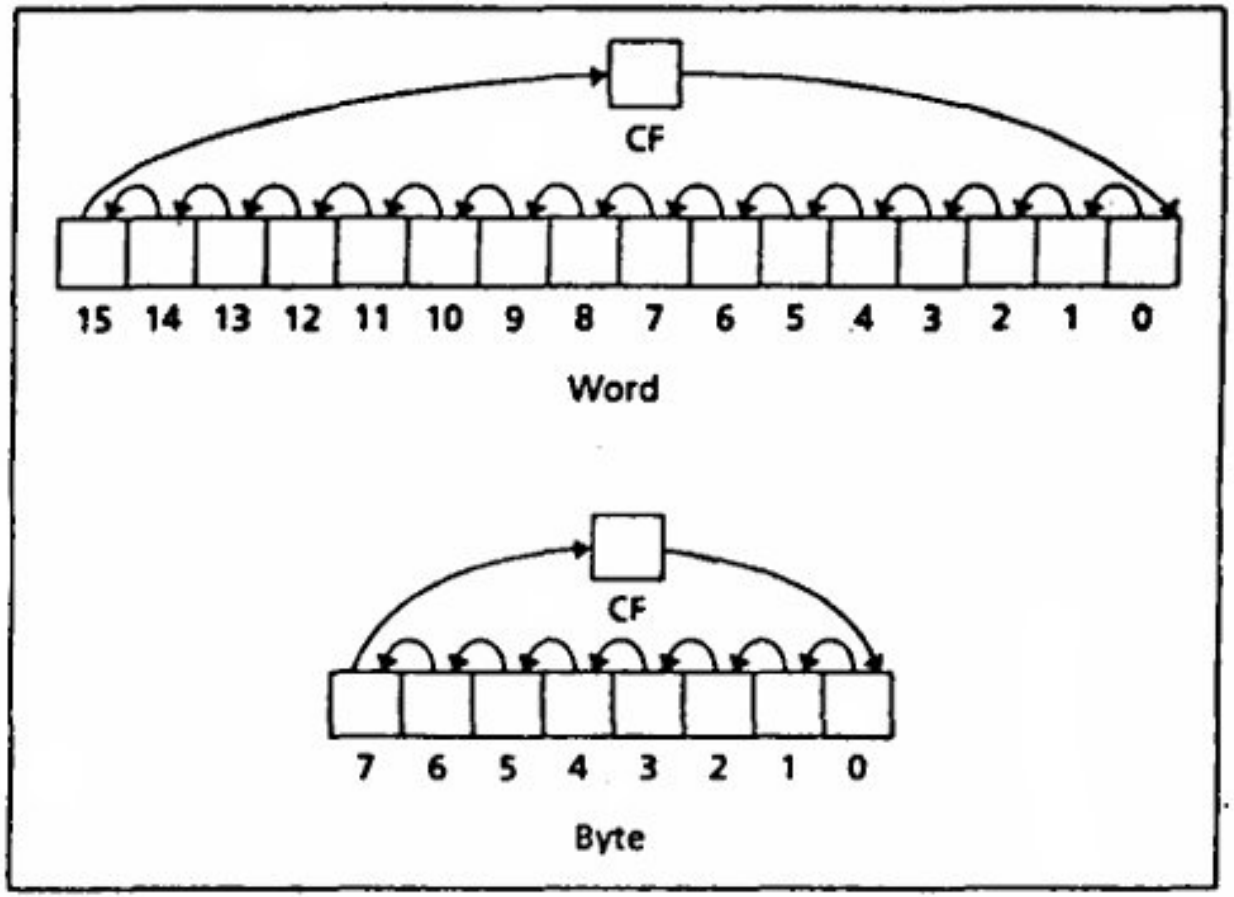
The instruction RCL (Rotate through Carry Left) shifts the bits of the destination to the left. The MSB is shifted into the CF, and the previous value of CF is shifted into the rightmost bit. In other words, RCL works like just like ROL, except that CF is part of the circle of bits being rotated. The syntax is,

```
RCL   destination,1
and
RCL   destination,CL
```

Rotate Instruction

Rotate Carry Left (RCL):

Figure 7.7 RCL



Rotate Instruction

Rotate Carry Right (RCR):

The instruction RCR (Rotate through Carry Right) works just like RCL, except that the bits are rotated to the light. The syntax is,

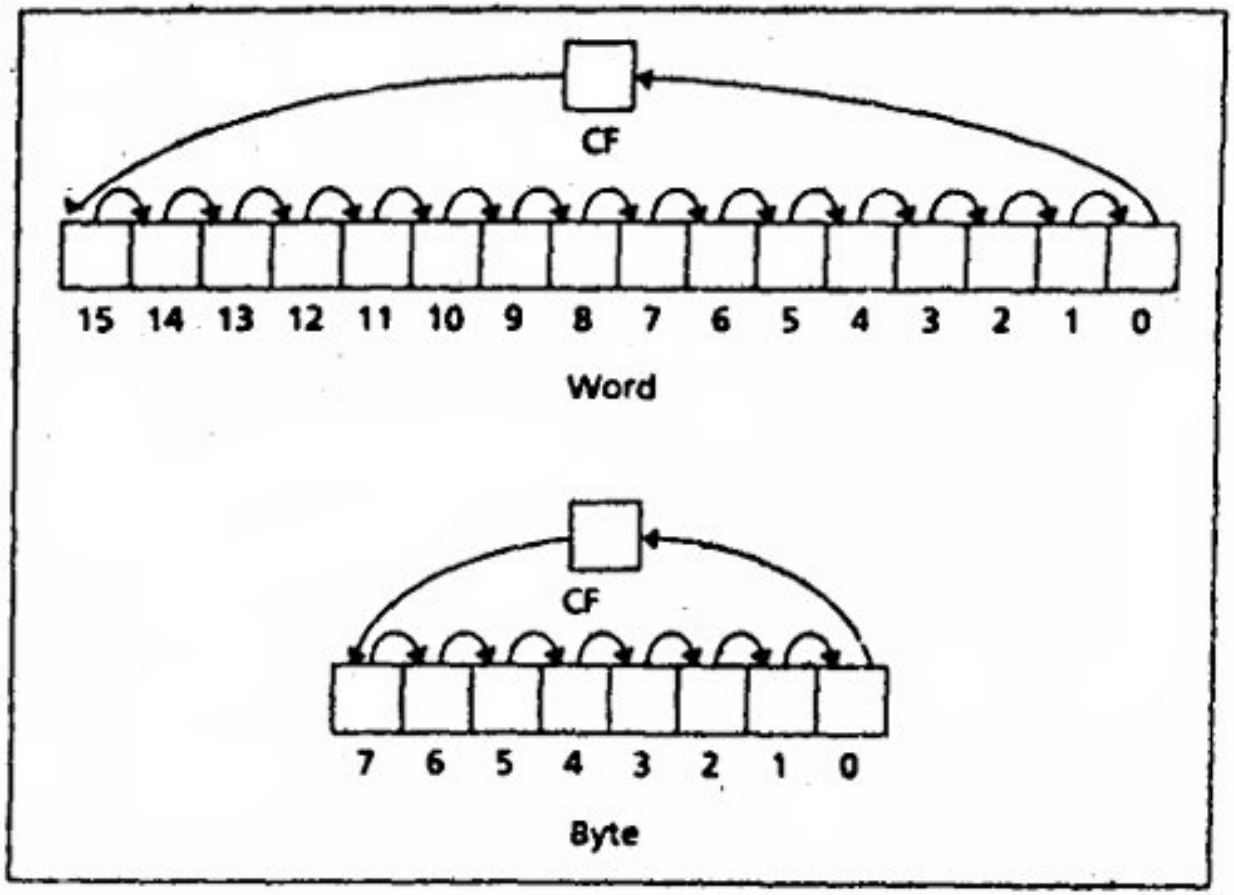
```
RCR   destination,1
and
RCR   destination,CL
```

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Rotate Instruction

Rotate Carry Right (RCR):

Figure 7.8 RCR



Rotate Instruction

Example 7.13 Suppose DH contains 8Ah, CF = 1, and CL contains 3. What are the values of DH and CF after the instruction RCR DH,CL is executed?

Solution:

	CF	DH
initial values	1	10001010
after 1 right rotation	0	11000101
after 2 right rotations	1	01100010
after 3 right rotations	0	10110001 = B1h

Effect of the rotate instructions on the flags

SF, PF, ZF reflect the result

AF is undefined

CF = last bit shifted out

OF = 1 if result changes sign on the last rotation

Rotate Instruction

1. ROL
2. RCL
3. ROR
4. RCR

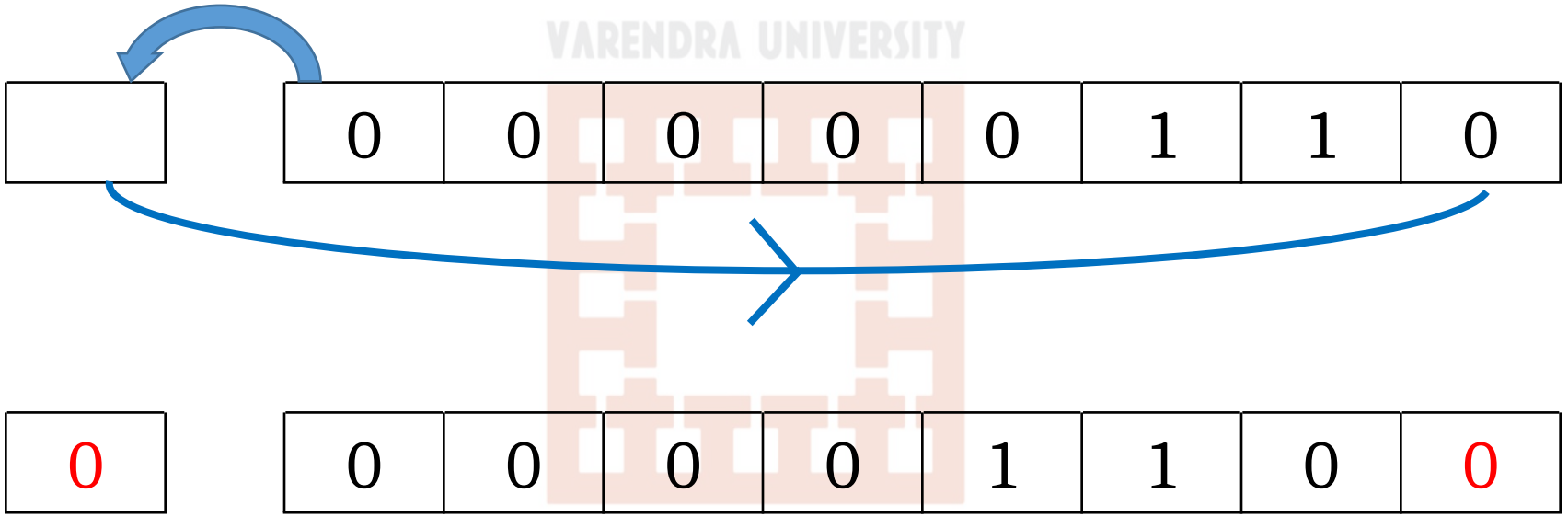


Binary Value is 6

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Rotate Instruction

ROL/RCL

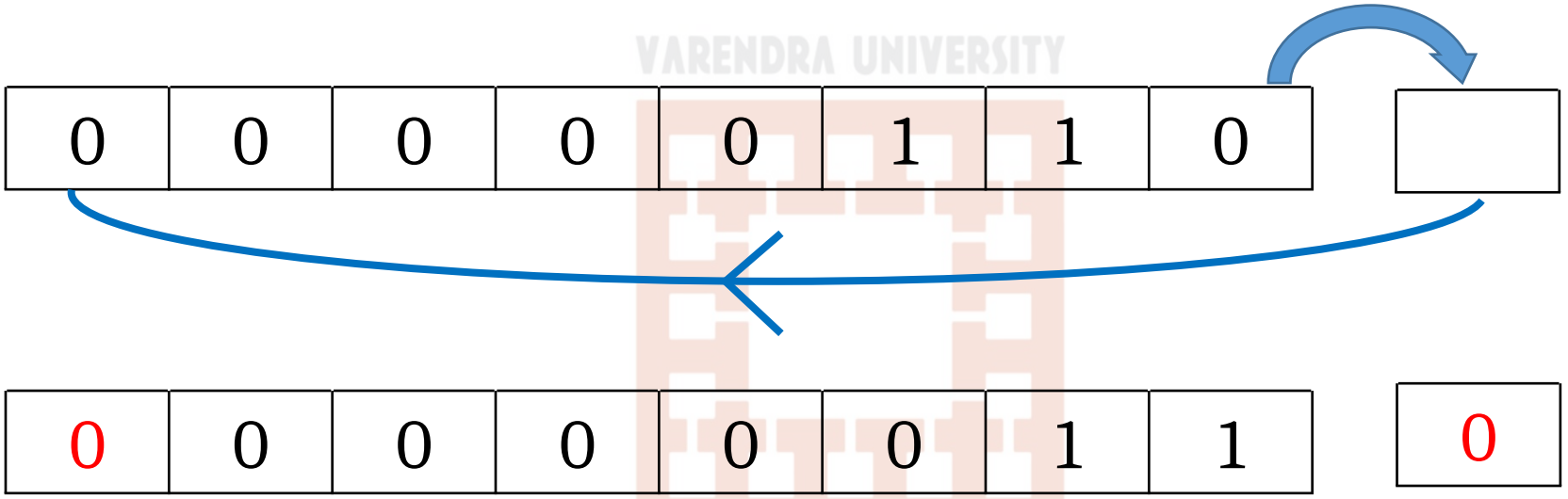


Binary Value is 12

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Rotate Instruction

ROR/RCR



Binary Value is 3

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Application of Shift and Rotate Instruction

- Reversing Bit Pattern.

The code is,

```
                MOV     CX,8        ;number of operations to do
REVERSE:
                SHL     AL,1        ;get a bit into CF
                RCR     BL,1        ;rotate it into BL
                LOOP    REVERSE     ;loop until done
                MOV     AL,BL       ;AL gets reversed pattern
```

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